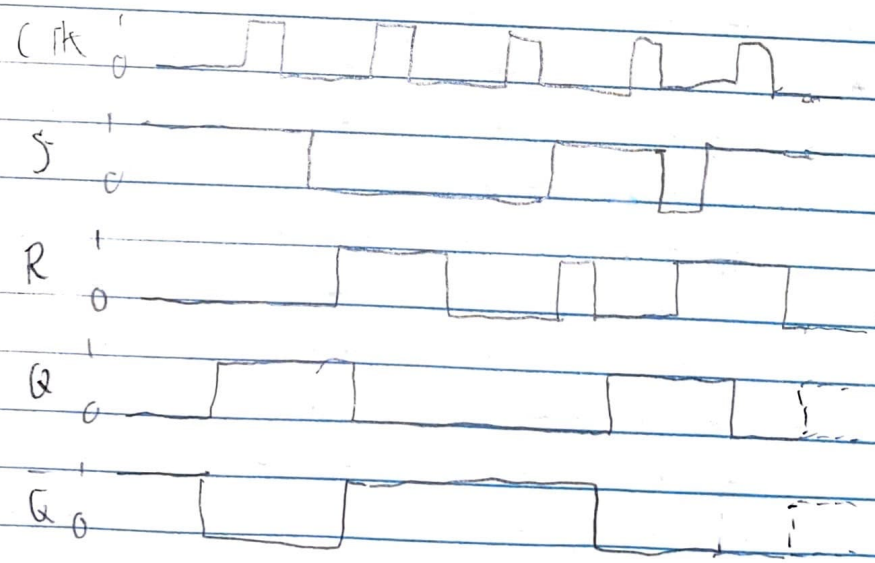
Dylan Treanting

1. A basic latch is circuit that changes output depending on input and stores previous data as output.

2. A gated latch is a latch that has a third input for it to work typically called enable

3.

| Clk | S | R | Q | $\overline{Q}$ |
|---|---|---|---|---|
| 0 | 0 | 0 | Q | $\overline{Q}$ |
| 0 | 0 | 1 | Q | $\overline{Q}$ |
| 0 | 1 | 0 | Q | $\overline{Q}$ |
| 0 | 1 | 1 | Q | $\overline{Q}$ |
| 1 | 0 | 0 | Q | $\overline{Q}$ |
| 1 | 0 | 1 | 0 | 1 |
| 1 | 1 | 0 | 1 | 0 |
| 1 | 1 | 1 | X | X |



$A(t+1) = A(t)x + B(t)x$

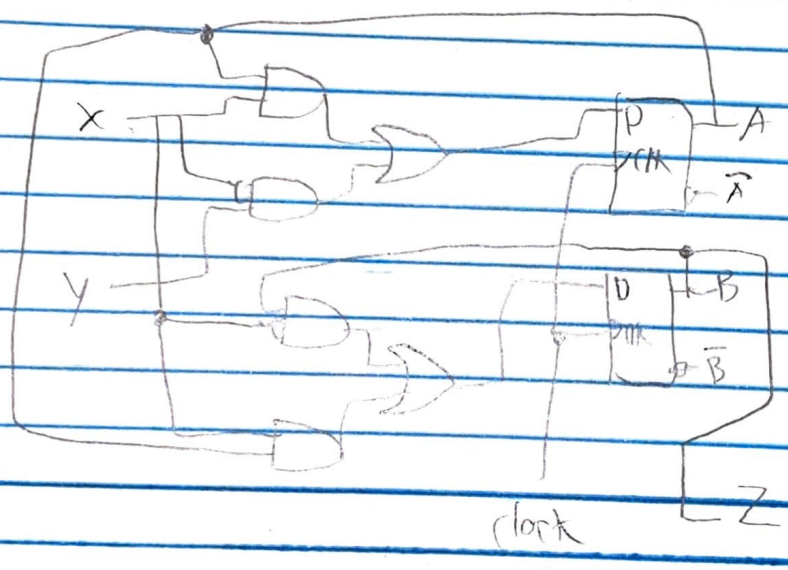
$B(t+1) = x\overline{A(t)}$

$y = \bar{x}(A+B)$

4.

| X | A(t) | B(t) | A(t+1) | B(t+1) | y |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 | 0 | 1 |
| 0 | 1 | 0 | 0 | 0 | 1 |
| 0 | 1 | 1 | 0 | 0 | 1 |
| 1 | 0 | 0 | 0 | 1 | 0 |
| 1 | 0 | 1 | 1 | 1 | 0 |
| 1 | 1 | 0 | 1 | 0 | 0 |
| 1 | 1 | 1 | 1 | 0 | 0 |

5.



| X(t) | y(t) | A(t) | B(t) | A(t+1) | B(t+1) | z(t) |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 1 | 0 | 1 | 1 |
| 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 0 | 1 | 1 |
| 0 | 1 | 0 | 0 | 1 | 0 | 0 |
| 0 | 1 | 0 | 1 | 1 | 1 | 1 |
| 0 | 1 | 1 | 0 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 0 | 1 | 1 | 0 |
| 1 | 0 | 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 0 | 0 | 0 | 0 | 0 |
| 1 | 1 | 0 | 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 0 | 1 | 1 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 |